Longest Common Subsequence

$$X = (x_1, x_2\ x_3 \cdots x_m)$$

$$Y = (y_1\ y_2\ y_3 \cdots y_n)$$

$$LCS(x_i, y_i) = \begin{cases} \phi & \text{if } i=0, j=0 \\ LCS(x_{i-1}, y_{j-1})\ x_i & \text{if } x_i = y_i,\ i,j>0 \\ \max[LCS(x_{i-1}, y_j), LCS[x_i, y_{j-1}]] & i,j>0,\ x_i \neq y_i \end{cases}$$

		A	G	C	A
	0	0	0	0	0
G	0	0	1	1	1
A	0	1	1	1	2
C	0	1	1	2	2

A C
G C
G A